TECTONIC IMPLICATIONS OF PERMO-TRIASSIC PALEOMAGNETIC RESULTS FROM NORTH AND SOUTH CHINA

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Abstract. Late Permian and Early Triassic paleomagnetic results of our own and others show that the North China Block (NCB), South China Block (SCB), and Siberia were not in their present positions relative to each other. Paleomagnetic poles for the NCB that derived from our studies of Permian rocks at several localities in Shanxi and Hebei provinces are concordant with poles from other studies of Permian rocks in Shanxi, Gansu, and Inner Mongolia. In addition, our poles for the Permian Emeishan Basalts and those from many other studies of the same formation over a sizeable portion of the SCB are consistent if we reinterpret the polarity of several of them. Early Triassic poles from both blocks are fairly close to the Late Permian ones, and greatly increase the areal extent covered by the data for the SCB. The equatorial paleolatitudes inferred by the data from both blocks are similar, but the mean declination for the SCB is rotated clockwise more than 60 degrees with respect to that for the NCB. To explain these results we propose a simple tectonic model in which initial collision of the blocks occurred in the Early Triassic near the eastern end of their boundary and progressed westward as the SCB rotated clockwise 67 degrees relative to the NCB. Eastward thinning and eventual disappearance of Triassic marine sediments on the northern margin of the SCB support this model. A simple alternative model involves sinistral movement on a transform fault that wraps around the SCB to the north and west, but evidence for the large Mesozoic displacement (2,500 km minimum) is presently lacking. Jurassic and Cretaceous poles for North and South China and for Siberia are overlapping, but no fold or reversal tests are available for the Jurassic data from China. Thus, the major movements between the three blocks were completed at least by the Late Cretaceous, perhaps during Jurassic time.

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Introduction

The eastern half of China is dominated by the North China Block (NCB) and South China Block (SCB), separated by a complex fold system (Qinling fold belt) that trends roughly east-west. Interpretation of the geology of this fold belt has led to a variety of estimates of the age of amalgamation of these two blocks. Klimetz [1983], Ji and Coney [1985], Sengor [1985], and Hsu et al. [1987] concluded that the SCB accreted to the NCB in Late Triassic to Early Jurassic time. On the other hand, Zhang et al. [1984], Mattauer et al. [1985], and Z. Q. Xu et al. [1986] proposed that the NCB and SCB were joined in mid-Paleozoic time, probably in the Devonian. In addition, some Chinese geologists interpret the collision to be Late Permian [Zhang and Zhang, 1982; Li et al., 1984, Wang, 1985], whereas others favor an Early to Middle Triassic (Indosinian orogeny) collision [Li and Tang, 1983; Wang et al., 1983; Zhu et al., 1984; Li et al., 1984; Yang et al., 1986; J. W. Xu et al., 1987].

Paleomagnetism can be most helpful in discriminating between these hypotheses, but results are few and have also been subject to conflicting interpretations [Wang et al., 1960; McElhinny et al., 1981; Chan et al., 1984; Lin et al., 1985; McElhinny, 1985]. A preliminary report of paleomagnetic results for Late Permian rocks of both the SCB and NCB and their implications has been published previously [Zhao and Coe, 1987]. In this paper we present the detailed account of our own results from the two blocks and summarize other up-to-date relevant paleomagnetic data, some of which were not previously available in western literature. By reinterpreting the polarity given in some previous studies we are able to make a satisfying synthesis of the paleomagnetic results for the widespread Emeishan basalts of the SCB and to suggest simple models for the relative rotation of the two blocks. We then compare and discuss available paleomagnetic data from other parts of eastern Eurasia, and address the tectonic implications.

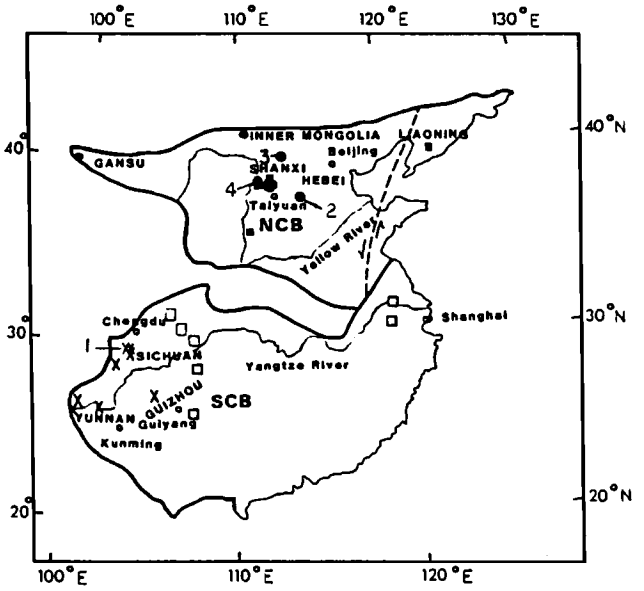


Fig. 1. Map of present location of the North China Block (NCB) and South China Block (SCB). Thin-line boundaries are shorelines (not block boundaries) and the broken line is the Tancheng-Lujiang fault. Sampling sites from the NCB and SCB are also indicated, with squares representing Triassic sites, and circles and crosses representing Permian sites. Sites of this study are denoted by numbers: 1. Emeishan, Sichuan; 2. Lincheng, Hebei; 3. Kouquan, Shanxi; 4. Jingle, Shanxi.

Geologic Setting and Paleomagnetic Sampling

Our sampling areas are confined to Hebei and Shanxi provinces on the NCB, and Sichuan province on the SCB (Figure 1). On the NCB, we sampled sedimentary sections of the Late Permian Shihezi Formation at three widely separated areas. In Shanxi province, a section over 800 m thick of Permian sandstone and siltstone is exposed in the Kouquan area (site Lat = 40.1 ^{o} N, Long = 113.2^{o} E) of the Datong Basin. In the Jingle area (site Lat = 38.6 ^{o}N, Long = 112.1 ^{o} E) more than 150 km to the southwest, the same formation is well exposed on both limbs of a syncline. In Hebei province, fresh exposures of a sandstone unit crop out along the banks of the Shaba ravine in Lincheng area (site Lat = 37.5 ^{o}N, Long = 114.4 ^{o}E). At all three localities the coal-bearing Shanxi Formation, which is Early Permian in age, can be easily recognized at the base of the sections. The strata we sampled are paleontologically dated as Late Permian [Chinese Academy of Geological Sciences, 1973; Shanxi Regional Stratigraphic Tables Compilation Team, 1979], and are mainly yellowish in color except at Jingle, where redbeds are common. Both local folding and regional deformation occurred in these two provinces in the Middle Jurassic and Early Cretaceous, corresponding in time to the Indosinian and Yanshanian orogenies, respectively [Chinese Academy of Geological Sciences, 1973; Hebei and Tianjin Regional Stratigraphic Tables Compilation Team, 1979; Shanxi Regional Stratigraphic Tables Compilation Team, 1979].

On the SCB, one of us (X.Z.) sampled 32 flows of the Late Permian Emeishan basalt formation at the type locality on the east limb of the Nubeishan anticline near Emei Mountain (site Lat = 29.6^{o} N, Long = 103.4^{o} E), Sichuan province in 1980. These flows span 212 meters of the section and are overlain by the Sawan Formation, which consists of reddish sandstone, siltstone, and greyish tuffaceous sandstone. Fossils found in this formation include Gigantopteris nicotianaefolia and Stigimaria permica (ficoides), which are indicators of Late Permian age [M. Q. Yao, personal communication, 1981]. The flows dip at 83° in a direction 336° (northwest). Regional geological data suggest that these basalts were folded after Cretaceous time [Sichuan Regional Stratigraphic Tables Compilation Team, 1978].

Sampling was carried out by using a portable drill, and samples were oriented with a magnetic compass. Whenever possible, solar orientations were also taken at each site to check the effect of local magnetic anomalies on magnetic orientaion. Solar and magnetic orientations were usually consistent within 1 degree. Bedding attitudes were measured and recorded directly at each sampling site.

Magnetic Measurement and Data Analysis

In the laboratory, two specimens on average were cut from each core. Specimens were demagnetized either in a nonmagnetic oven at successive temperature steps and then cooled in a quasi-zero (< 20 gammas) field space or by use of an increasing alternating field. Both types of demagnetization were performed with Schonstedt equipment. Magnetic measurements on Emeishan basalts from the SCB were done with an astatic magnetometer in Beijing, China. Measurements on sandstone from the NCB were performed with a Schonstedt spinner magnetometer and a 2G cryogenic magnetometer in the paleomagnetic laboratory of UCSC. Magnetizations were determined by the principal component analysis method [Kirschvink, 1980] and the method of combined data from the remagnetization plane and least-squares-best-fit line, as described by McFadden and McElhinny [1988]. Confidence limits were calculated with Fisher [1953] statistics, and site mean directions of all demagnetized data were derived by giving unit weight to each mean sample direction.

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Paleomagnetic Results From the SCB

Most paleomagnetic data of Permian age for the SCB are derived from the Emeishan Basalt Formation, which is distributed widely throughout southwest China. The results of our own study, published previously as a masters thesis [Zhao, 1981] in Beijing, are presented briefly in Table 1. Both alternating field and thermal stepwise demagnetization experiments were conducted on the Emeishan basalt samples. In most cases, both types of treatments were equally successful. A stable (characteristic) component of magnetization was isolated for about 90 percent of the samples that decayed univectorially toward the origin (Figure 2). The directions of this stable component display a single polarity and cluster well about a mean with D = 357.7°, I = 40.2°, N = 44 samples, alpha-95 = 3.6°, k = 37.4 in geographic coordinates, and D = 19.6°, I = -11.6°, alpha-95 = 3.6°, k = 37.4 in stratigraphic coordinates. For reasons that will become apparent below, we interpret these data as primary directions acquired in a field of normal polarity.

The results of four other studies [McElhinny et al., 1981; Chan et al., 1984; Zhang, 1984; and Liu et al., 1987] of the Emeishan Basalts in the same area have revealed very similar northeast and upward paleomagnetic directions in stratigraphic coordinates (Table 1). Application of the F-test of Watson [1965] reveals that these directions are all indistinguishable within 95% confidence from each other and from our own. The data for one of these studies [McElhinny et al., 1981] pass the fold test at 99% confidence, which means that the magnetization predates the folding in Cretaceous, a necessary (but not sufficient) requirement for the directions to be primary. Although all our samples were from the east limb of the Nubeishan anticline, the data of six flows from limb 3 in McElhinny et al.'s [1981] study are from the same basalt on the other limb of the fold, which

TABLE 1. Summary of Late Permian and Triassic Paleomagnetic Results on Rocks From the SCB and the NCB

Study	Age	Rock Type & Locality	SLat	SLong	Ns	Nf	P	Dec	Inc	a95	k	PLAT	PLONG	A95
South China Block (SCB)														
This study	P_2	Emeishan Basalt, Sichuan	29.6	103.4	44	32	N	19.6	-11.6	3.6	37.4	49.7	252.0	2.8
McElhinny et al. (1981)	P_2	Emeishan Basalt, Sichuan	29.6	103.4	?	22*	N	18.4	-6.5	6.2	26.4	54.0	252.0	3.8#
Chan et al. (1984)	P_2	Emeishan Basalt, Sichuan	29.6	103.4	7	?	N	22.9	2.2	19.5	10.5	54.1	241.8	15.9
Zhang et al. (1984)	P_2	Emeishan Basalt, Sichuan	29.6	103.4	72	20	N	21.4	-7.6	3.3	26.8	50.7	248.3	2.0#
Liu et al. (1987)	P_2	Emeishan Basalt, Sichuan	29.7	103.5	16	?	N	38.4	-11.8	5.1	53.0	38.5	231.6	3.1#
	P_2	Emeishan Basalt, Sichuan	28.1	102.9	19	?	N	18.1	-4.9	7.0	24.0	58.9	246.1	4.3#
	P_2	Emeishan Basalt, Sichuan	27.6	101.9	17	?	N	7.8	-3.9	10.7	12.0	63.5	264.3	6.6#
Mean of Liu et al. study		N = 3 Localities					N	21.3	-7.0	24.8	25.8	54.4	244.2	25.3
Zhou et al. (1986)	P_2	Emeishan Basalt, Sichuan	28.3	103.0	20	2	N	17.9	-5.9	—	14.8	54.3	251.3	—
	P_2	Emeishan Basalt, Sichuan	26.8	101.8	31	3	M	235.9	15.8	12.2	102.8	25.6	216.5	8.7
Mean of Zhou et al. study						5	M	40.8	-12.5	23.6	11.5	37.9	225.6	23.2
Huang et al. (1986)	P_2	Emeishan Basalt, Sichuan	26.7	102.9	7	7	M	23.0	-4.5	10.1	37.0	53.5	241.8	8.5
	P_2	Emeishan Basalt, Yunnan	26.1	103.1	8	8	M	30.8	5.7	25.0	5.9	53.3	224.8	16.5
	P_2	Emeishan Basalt, Yunnan	25.9	100.6	3	3	R	242.0	1.3	24.8	25.9	-24.6	24.4	18.1
Mean of Huang et al. study						18	M	32.9	0.3	12.5	8.7	49.5	226.2	9.8
Lin et al. (1985)	P_2	Emeishan Basalt, Guizhou	26.4	105.7	5	1	R	224.5	30.3	12.0	41.0	-29.3	55.3	12.0
Chan et al. (1984)	T_1	Limestone, Guizhou	26.6	106.5	8		R	215.7	-26.1	5.7	95.0	-54.6	29.1	3.0#
Opdyke et al. (1986)	T_1	Shale & Limestone, Guizhou	28.6	106.9	47		N	40.0	13.1	10.9	23.1	46.3	219.1	2.6#
	T_2+	Limestone, Jiangsu	32.0	119.0	23		R	224.8	-24.4	9.3	11.5	-44.8	43.6	4.9#
Lin et al. (1985)	T_1	Limestone, Zhejiang	31.0	119.8	42		N	44.8	44.9	6.8	38.0	50.7	204.8	1.3#
Zhang et al. (1987)	T_1	Marl, Hechuan, Sichuan	30.0	106.3	80		N	50.2	18.0	3.5	21.3	38.9	209.3	2.0#
	T_1	Marl, Guangyuan, Sichuan	32.4	105.9	53		N	39.5	18.4	4.9	17.0	46.7	219.6	2.8#
	T_1	Limestone, Chongqing	29.6	106.4	16		N	30.6	8.7	9.3	16.6	51.6	231.7	5.7#
Heller et al. (1988)	T_1	Limestone, Guangyuan	32.4	105.5	114		N	35.9	12.8	5.3	7.1	47.7	225.5	4.2
Grand average for Permian: method 1[a]		N = 5										45.0	235.2	12.2
method 2[b]		N = 8										47.9	239.9	8.0
method 3[c]		N = 6										50.0	240.6	8.4
Grand average for Permian & Triassic:		N = 14										49.5	227.3	5.7
Grand average for Triassic:		N = 8										48.0	217.8	5.3

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TABLE 1. (continued)

Study	Age	Rock Type & Locality	SLat	SLong	Ns	Nf	P	Dec	Inc	a95	k	PLAT	PLONG	A95
North China Block (NCB)														
This study	P_2	Sandstone, Lincheng, Hebei	37.5	114.4	12		M	326.3	34.3	6.9	42.0	55.2	1.4	6.3
This study	P_2	Redbeds, Jingle, Shanxi	38.6	112.1	33		M	310.7	20.0	7.5	14.2	37.9	2.9	6.9
This study	P_2	Sandstone, Kouquan, Shanxi	40.1	113.2	19		M	336.8	23.0	9.0	15.2	55.2	335.7	4.7
McElhinny et al. (1981)	P_2	Redbeds, Taiyuan, Shanxi	37.8	112.4	12		R	138.4	-21.5	6.9	40.3	-44.0	178.5	3.9#
Zhang et al. (1984)	P_2	Redbeds, Taiyuan, Shanxi	37.8	112.4	82		R	141.8	-27.9	3.9	17.4	-49.2	178.5	2.0#
J. Lin (1984)	P_2	Redbeds, Taiyuan, Shanxi	37.8	112.3	10		R	129.3	-23.0	?	?	-38.1	186.3	?
W. Lin et al. (1984)	P_2	Redbeds, Taiyuan, Shanxi	37.9	113.8	8		M	325.3	40.1	10.9	49.8	56.7	6.2	11.5
Coe et al.(1987)	P_2	Redbeds, Dahuanggou, Gansu	39.8	97.8	30		R	133.8	-29.5	4.5	35.0	-43.9	171.3	4.1
H. Liu (1987)	P_2	Sandstone, Inner Mongonlia	42.3	110.2	50		N	327.1	40.6	?	?	56.7	355.6	?
J. Lin (1984)	T_1	Redbeds, Taiyuan, Shanxi	37.8	112.3	7		N	315.0	31.2	15.9	15.0	45.4	6.8	7.7#
Zhang et al. (1984)	T_1	Redbeds, Taiyuan, Shanxi	37.8	112.4	56		N	328.5	31.5	4.8	17.0	55.5	354.1	2.3#
X. Tan (1988)	T_1	Redbeds, Jingle, Shanxi	38.5	112.1	33		M	328.9	40.0	3.7	44.4	60.3	3.5	1.6#
	T_1	Redbeds, Ji county, Shanxi	36.2	110.8	5		M	329.8	33.1	9.8	40.4	56.8	352.7	5.8#
J. Lin (1984)	T_1	Redbeds, Benxi, Liaoning	41.3	123.7	28		R	280.7	35.2	21.7	10.5	21.8	42.7[$]	19.7
Grand average for Permian:		N = 6 (Studies in Taiyuan grouped as one unit)										49.7	355.4	8.3
Grand average for Triassic:		N = 4 (Study from Benxi omitted, see text)										52.7	358.7	12.0
Grand average for Permian & Triassic:		N = 10										50.7	356.5	5.8
		N = 7 (Triassic studies grouped as one unit)										50.1	355.9	6.9

Age: approximate age of rocks studied, based on paleontological and geological evidence;
P_2 = Late Permian, T_1 = Early Triassic, T_2 = Middle Triassic;
+: age revised downward from Late Triassic according to Weng and Wang [1981];
SLat: site latitude; SLong: site longitude; Ns: number of samples used;
Nf: number of flows sampled; P: polarity (N-normal, R-reversal, M-mixed); Dec: declination;
Inc: inclination (directions corrected for tilt);
a95: semiangle of the cone of 95% confidence; k: precision parameter for directions; PLAT: latitude of pole;
PLONG: longitude of pole;
A95: circle of 95% confidence limit of pole, #: A95 estimated by method in Cox [1970] & Coe et al. [1985];
*Includes 2 sandstone sites;
[$]Recalculated on the basis of 6 site means given by Lin [Lin 1984, Table 1, p. 16];
[a]Averaged by regions, N = 5; [b]Averaged by studies, N = 8;
[c]Averaged by studies, Chan et al. [1984] and Lin et al. [1985] omitted because of small number of samples, N = 6.
Underlined data are recalculated from original publication.

allows us to perform a "hybrid" fold test on our data. We used the fold test proposed by McFadden and Jones [1981], which compares the lengths of the resultant vectors from each limb of the fold to test the probability that they have a common mean direction. The directions of the two limbs are different at less than 95% confidence, suggesting that they cannot be considered to not share a common direction. In all four previous studies in the Emei Mountain area of Sichuan province, the polarity assigned is opposite to our interpretation; that is, reversed. The reason for the reversed polarity choice is that the age of the Emeishan Basalt Formation was thought to lie within the late Paleozoic (Kiaman) Reversed Interval [Irving and Parry, 1963]. Assignment of reversed polarity to these paleomagnetic directions with northeast declinations, however, implies that the area has undergone an azimuthal rotation since Permian time of about 160 degrees. McElhinny et al. [1981] originally interpreted this as a rotation of the entire SCB. Chan et al. [1984], however, pointed out that the area sampled might have undergone a local rotation because it lies within the Lungmenshan thrust zone.

The interpretation of paleomagnetic results from the Emeishan Basalt Formation was renewed when recent studies of the Emeishan Basalts in other regions revealed that some flows have directions with southwesterly declinations and shallow downward inclinations. Lin et al. [1985] reported a paleomagnetic pole position corresponding to such a direction derived from five specimens of a single lava flow [Lin, 1984, p. 184] in Guizhou province (Table 1). Although the accuracy of this pole must be questioned because a single flow cannot be expected to average out secular variation, we agree with their interpretation that this direction represents reversed primary magnetization. They attributed the discrepancy between their direction and the earlier result of McElhinny et al. [1981] to secondary remagnetization or local

rotation of the Emei Mountain region. In a later paper, McElhinny [1985] reinterpreted their 1981 results by pointing out structural evidence that the area they sampled may have rotated about 110 degrees, suggesting that the inclination is correct but the declination does not refer to the SCB as a whole. Subsequently, studies by Huang et al. [1986] and Zhou et al. [1986] of Emeishan Basalt Formation from southern Sichuan and Yunnan provinces report directions for sequences of flows with both northeasterly and southwesterly

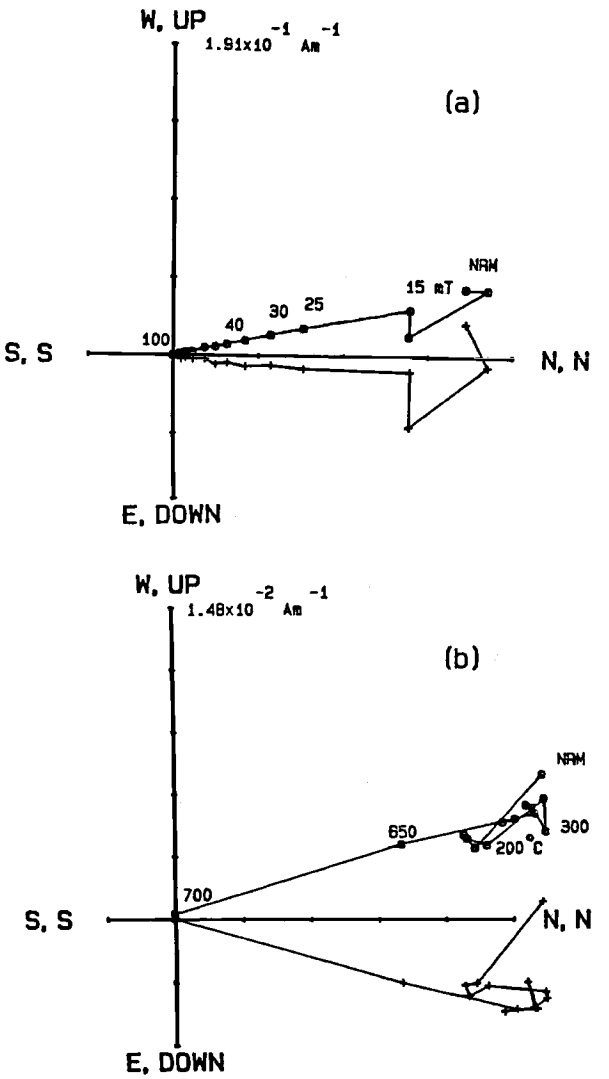


Fig. 2. Representative orthogonal demagnetization plots of the Emeishan basalt samples in this study: (a) AF demagnetization results; (b) thermal demagnetization results. Directions are plotted in stratigraphic coordinates. Crosses represent vector endpoints projected onto horizontal plane; solid circles, endpoints projected onto the north-south vertical plane.

declinations. Particularly diagnostic is the latter, in which the upper flows have northeasterly and upward directions, and the lowest flow has a southwesterly and downward direction. The underlying, Late Permian bauxite beds of the Liangshan Formation also have southwesterly declinations. Thus, the simplest interpretation for all the results is that the Emeishan Basalt Formation does not lie exclusively within the Kiaman Reversed Interval, and that the results record both normal (NE declination) and reversed (SW declination) polarities of the ancient field [Zhao and Coe, 1987]. This interpretation is also one of three possible explanations of Emeishan data offered by Huang et al. [1986].

Recently published radiometric dates from the Emeishan Basalt support this interpretation. Yuan et al. [1985] found an age range of 219 to 253 Ma for the basalt, with a tendency toward older ages to the south. They cite a mean age of 235 Ma for flows from Sichuan province, which roughly corresponds to the upper limit of the Kiaman Reversed Interval [Irving and Pullaiah, 1976; Khramov and Rodionov, 1981; Harland et al., 1982; Palmer, 1983; Opdyke, 1986]. If these ages are accurate, it is not surprising that both normal and reversed polarities are found and that, in particular, the polarity recorded by the flows at the type locality near Emei Mountain, the northernmost area sampled, is normal.

In this way, the results of all the previous studies are acceptable and the need for exceptionally large azimuthal rotations can be eliminated. Results for the various studies of the Emeishan Basalt are listed in Table 1. Some of them were obtained by recalculating data from the original paper, as indicated in Table 1. In general, these directions agree very well with each other and, as discussed above, some of them pass fold tests or display reversals. For these reasons, and because of the wide distribution of sampling sites, we believe that these are primary directions suitable for calculating a Late Permian pole position for the SCB. We have done this in three different ways: (1) averaging by regions (N = 5); (2) averaging by studies (all studies, N = 8); (3) averaging by selected studies (omitting two studies of small numbers of samples as indicated in Table 1, N = 6). All three methods yield similar results, but we prefer the third method because the mean directions used to obtain the pole are more nearly comparable in terms of number of samples and flows (see Figure 3).

Paleomagnetic Results From the NCB

Demagnetization Properties Common to Three Areas

Four thick sections of Late Permian sedimentary rock units have been sampled from the Kouquan and Jingle areas in Shanxi and the Lincheng area in Hebei provinces. In general, rocks from these

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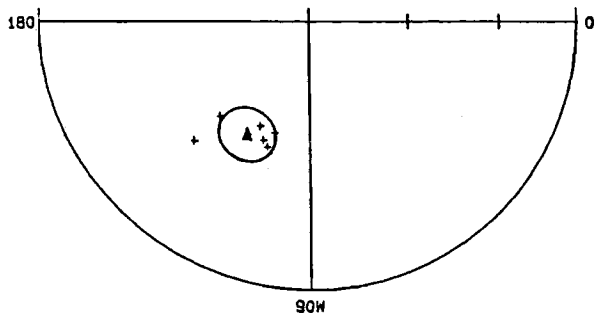


Fig. 3. Equal-area stereographic projection of averaged virtual geomagnetic poles and the 95 percent confidence limit for six studies of the Late Permian Emeishan Basalt Formation (Table 1). Crosses represent site mean VGP's for individual studies; corresponding A95 circles have been omitted for clarity. Large triangle is the mean of the six VGPs with the A95 (solid circle).

four sections exhibited very similar magnetic behavior in the laboratory. Both thermal and AF demagnetization techniques were applied but in most cases only thermal proved to be effective in cleaning. In the initial stages of demagnetization, a component with a direction close to the present field was removed. Further progressive thermal demagnetization by about 500° C removed a large component that is of dual polarity. This component, as discussed in detail by Zhao [1987], represents a regional chemical remagnetization that occurred in late Mesozoic time during the Yanshanian orogeny. For the purpose of convenience, this component is hereafter termed the Mesozoic overprint component.

In about 65 percent of the samples from each site, a higher unblocking temperature component is isolated. This higher unblocking temperature component has southeasterly or northwesterly declinations and shallow upward or downward inclinations, respectively. F-ratio criteria indicate that the two directions of the high unblocking temperature component are not significantly different at 95% confidence level if one polarity is inverted to the other. In each of the five sites, we also found evidence for this component in the demagnetization planes of the remaining 35 percent of the samples. We analyze our data using the improved methodology of joint demagnetization line and plane by McFadden and McElhinny [1988]. Detailed accounts of demagnetization behavior for each area are given below. The paleomagnetic results of this analysis are summarized in Table 1.

The Lincheng Area of Hebei Province

The Shihezi Formation site was sampled along the banks of the Shaba ravine of Linchen county, which is about 50 km southeast of Jingxin. This rock unit consists of a thick section of tan to orange, fine- to medium-grained sandstones. The beds overlie the Early Permian Shanxi Formation that contains coal and includes the fossils Corddites schenkii and Rhipidopsis lobata, which are indicative of a Late Permian age [Hebei and Tianjin Regional Stratigraphic Tables Compilation Team, 1979; Yang et al., 1986].

The initial NRM intensities for these sandstones vary from 3 x 10^{-4} to 8 x 10^{-3} A/m. Upon progressive thermal demagnetization to 600° C, the dominant Mesozoic overprint component, which is of dual polarity, was removed from most of the samples (Figure 4). In the six samples that were from the lower part of the section, the characteristic component is clearly visible in the demagnetization paths after 600 °C thermal demagnetization (Figure 4). This higher unblocking-temperature characteristic component is also present in six samples from the middle section, but the direction can be convincingly determined only from their demagnetization planes. Both normal (northwesterly and downward) and reversed (southeasterly and upward) polarities of the characteristic component are present, and the two polarities are close to antiparallel. By searching our field notes, we noted that the reversely magnetized samples are mainly confined to beds in the lower part of the section. The mean direction of the characteristic component for the six samples with linear demagnetization lines is D = 316.7°, I = -0.9°, alpha-95 = 12.4°, and k = 30.0 in geographic coordinates and D = 324.7°, I = 34.4°, alpha-95 = 11.3°, k = 36.4 in stratigraphic coordinates. The mean for the combined data set (six line and six plane data, respectively) is D = 317.9°, I = -0.4°, alpha-95 = 8.0°, k = 32.2 in geographic coordinates, and D = 326.3°, I = 34.3°, alpha-95 = 6.9°, k = 42.0 in stratigraphic coordinates, obtained by using the new method of McFadden and McElhinny [1988].

The Kouquan Area of Shanxi Province

The Late Permian sandstone of the Shihezi Formation is well exposed in the Kouquan area of the Datong basin, Shanxi province. Like its counterpart in the Lincheng area of Hebei province, the sandstone section is yellowish in color and thick bedded. Reported fossils in the section include Saporfaea, Protopinus sp., and Rhipidopsis, which are indicative of a Late Permian age [Chinese Academy of Geological Sciences, 1973]. The familiar Mesozoic overprint component is seen in 20 samples, with both polarities. The characteristic component is isolated in seven samples, also with two polarities. Evidence of the characteristic component is also found in remagnetization planes of 12 samples. In addition, eight samples were rejected for a variety of reasons, including evidence for having been severely affected by lightning and error during collecting in the

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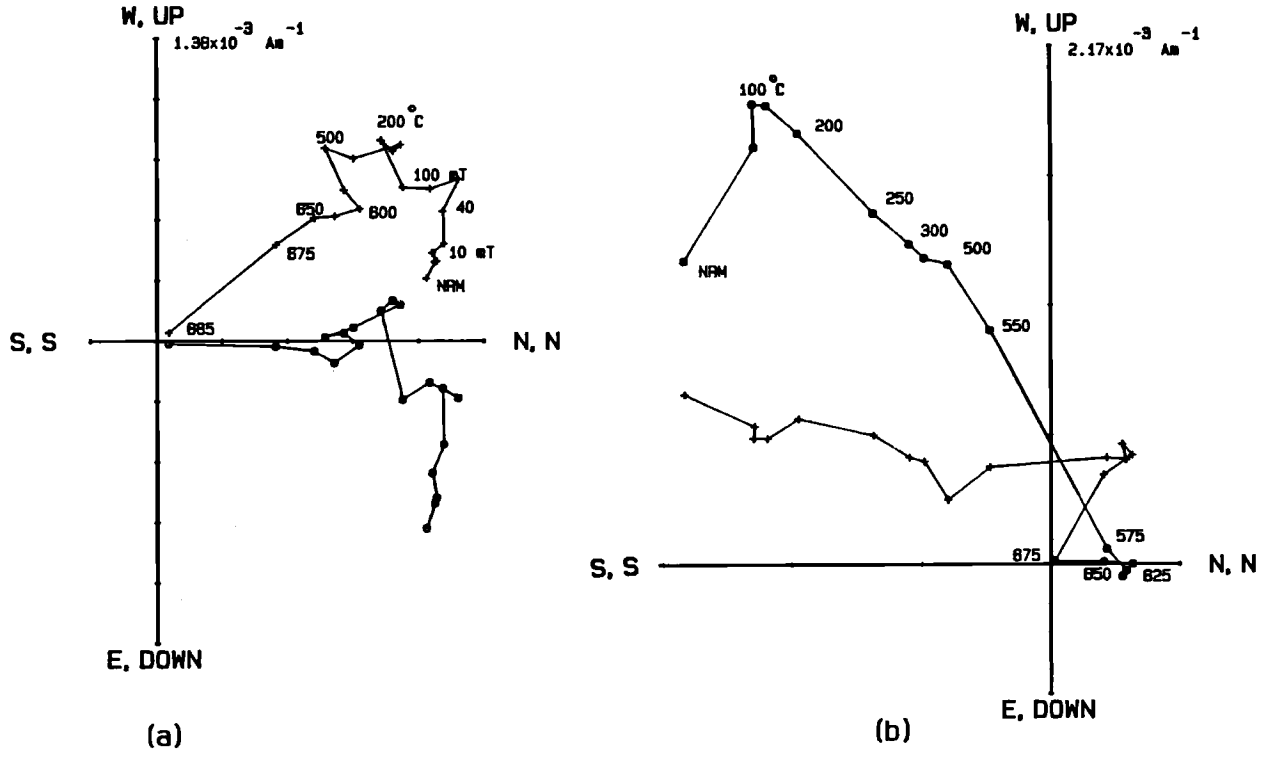


Fig. 4. Representative vector plots of demagnetizations on samples from the Lincheng area of Hebei province in the NCB. (a) AF and thermal demagnetization results of sample 83X0986B, showing the normal Mesozoic overprint component and the characteristic component; (b) thermal demagnetization on sample 83X0987A initially removed a large reversed Mesozoic overprint component, and then revealed the characteristic component. Both samples plotted in geographic coordinates. Crosses represent vector endpoints projected onto horizontal plane; solid circles, endpoints projected onto the north-south vertical plane.

field. The in situ mean direction for the seven samples with univectorial decay lines is D = 330.6 ^{o}, I = 39.9^{o}, alpha-95 = 8.9 ^{o}, and k = 47.2. After tectonic correction, the mean is D = 334.1 ^{o}, I = 13.9^{o}, alpha-95 = 8.1 ^{o}, and k = 56.2. The mean direction determined from the 12 planes by using the method of McFadden and McElhinny [1988] is similar to that of line data at 95% confidence level, using the test of Watson [1965]. The final direction found by combining the two data sources is D = 332.5^{o}, I = 48.8^{o}, alpha-95 = 9.1 ^{o}, k = 14.8 in geographic coordinates, and D = 336.8^{o}, I = 23.0^{o}, alpha-95 = 9.0^{o}, k = 15.2 in stratigraphic coordinates.

The Jingle Area of Shanxi Province

The Late Permian Shihezi Formation is well exposed along the two limbs of the syncline in the Jingle area of Shanxi province. Two sandstone sites were sampled. Rocks from the west limb are mainly yellowish in color, but local redbeds are found interbedded with massive layers of yellowish-green sandstone. In the east limb, however, the rocks are almost all redbeds. The NRM intensities for the yellow sandstone are typically on the order of 10^{-4} A/m, whereas for the redbeds, 10^{-3} A/m. At both sites, the Mesozoic overprint component is present in almost every sample and has normal polarity. Thermal demagnetization up to 600 ^{o}C is generally required to remove this overprint. In the site from the west limb, the characteristic component can be identified from the demagnetization plots of seven samples, with a mean in situ direction of D = 318.3^{o}, I = -45.0^{o}, alpha-95 = 16.0 ^{o}, k = 15.2, and D = 309.0 ^{o}, I = 12.5 ^{o}, alpha-95 = 15.2 ^{o}, k = 16.8 in stratigraphic coordinates. The characteristic component also exists in three other samples as revealed by their remagnetization planes. The mean for combined line and plane data is D = 315.4 ^{o}, I = -42.5 ^{o}, alpha-95 = 13.8 ^{o}, k = 13.5 in geographic coordinates, and D = 313.8 ^{o}, I =

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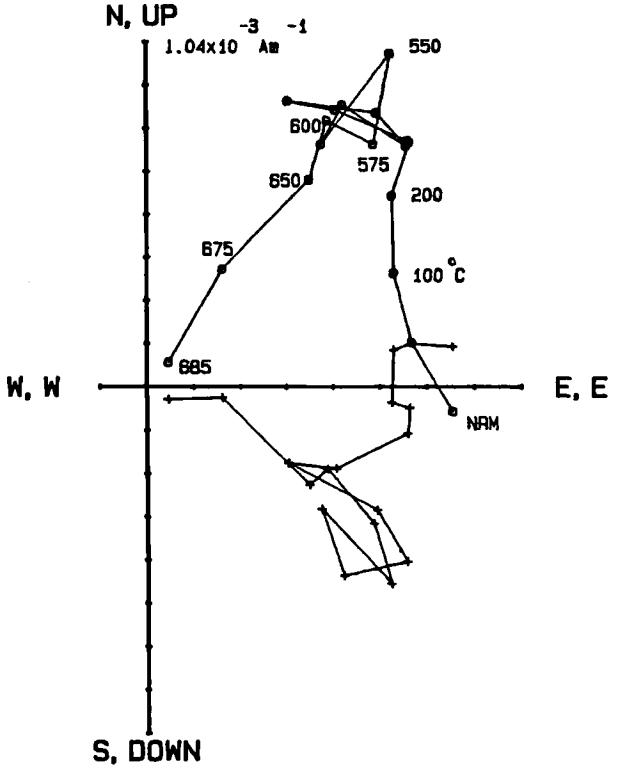


Fig. 5. Representative vector plot of thermal demagnetization on sample 83X1108A from the Jingle area in Shanxi province on the NCB. Directions are plotted in geographic coordinates, showing the normal overprint and the characteristic components. Crosses and solid circles have same meaning as Fig. 2.

24.2 °, alpha-95 = 8.4 °, and k = 13.9 in stratigraphic coordinates.

As expected, rocks from the site on the east limb exhibited demagnetization behavior similar to their counterparts on the west limb. The normal Mesozoic overprint component is present in all samples, but can be generally removed by 550° C thermal demagnetization (see Figure 5). In this site, 21 samples provide demagnetization lines for determination of the characteristic component, and two samples yield well defined remagnetization planes. The 21 line data group at a mean with D = 324.3° , I = 41.7°, alpha-95 = 8.9 °, k = 13.8 in geographic coordinates, and D = 313.1°, I = 23.4° , alpha-95 = 8.9°, and k = 13.8 in stratigraphic coordinates. The mean for both data sources is D = 326.1°, I = 42.2°, alpha-95 = 8.4 °, k = 13.9 in geographic coordinates, and D = 313.8 °, I = 24.2 °, alpha-95 = 8.4°, k = 13.9 in stratigraphic coordinates.

The Mesozoic overprint component from the two limbs of the fold has a mean direction of D = 3.0° , I = 55.0°, N = 56 samples, alpha-95 = 4.0 °, k = 23.7 in geographic coordinates, and D = 10.4 °, I = 47.5 °, alpha-95 = 10.7 °, k = 4.1 in stratigraphic coordinates (Table 2). The fold test is negative at more than the 99% confidence level, as indicated by a factor of 5.8 times decrease in Fisher's precision parameter k. The negative fold test demonstrates that the Mesozoic overprint was imposed after the folding in Middle Jurassic.

The characteristic component from the two sites, on the other hand, passes the fold test at more than the 95% confidence level. Application of the fold test of McFadden and Jones [1981] indicates that the hypothesis of a common true mean direction of samples in geographic coordinates may be rejected, but the hypothesis of a common true mean direction in stratigraphic coordinates cannot be rejected. An incremental fold test further suggests that the fold test is entirely positive. Thus, the remanent magnetization at these two sites was acquired before folding in Middle Jurassic time. The presence of closely antiparallel directions at each site provides further evidence for ancient magnetization. The data at hand therefore suggest that the characteristic component is

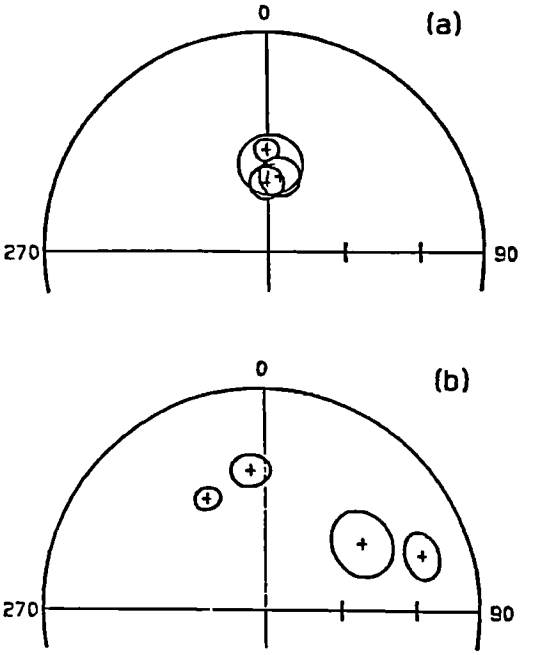


Fig. 6. Equal-area projection of mean directions (with alpha-95 confidence cones) of the Mesozoic overprint component of the four sites (a) before and (b) after structural corrections, showing failure of the regional fold test.

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primary or a very early phase of chemical remanent magnetization acquired during diagenesis shortly after deposition.

Discussion of Paleomagnetic Results From the Three Localities

The main paleomagnetic signatures from these three sampled localities in Hebei and Shanxi provinces are that both the Mesozoic overprint component and a higher unblocking temperature component can be identified in all four sites, which are spread over a wide area of the NCB. The Mesozoic overprint component fails both local and regional fold tests at more than 95% confidence (Table 2 and Figure 6). The higher unblocking temperature component, on the other hand, passes local and regional fold tests at more than 95% confidence level (Table 3 and Figure 7). Thus, the remanent magnetization at these three localities was acquired before folding and regional deformation in the Mesozoic. Moreover, both normal and reversed directions are found that are closely antiparallel. The ancient magnetization is carried most of the time by hematite, but indications of magnetite are also present as suggested by lower unblocking temperatures and coercivity forces, and by a steep initial increase and by saturation of the slopes in IRM acquisition curves at fields of 0.3 T or less. The successful fold and reversal tests, and the fact that this component is carried by magnetic minerals that are capable of preserving the early remanent magnetization over a long period of time, strongly suggest that the magnetization is primary or a CRM acquired very soon after deposition of these Late Permian rocks. The mean paleomagnetic pole position for all three localities in Hebei and Shanxi is Long = 357.6° E, Lat = 44.6° N, A_{95} = 15.6°, and K = 35.5.

Comparison with Other Permian Paleomagnetic Data From the NCB

Six other Permian paleomagnetic results are available for the NCB [McElhinny et al., 1981; J. Lin, 1984; J. Lin et al., 1985; Zhang, 1984; Liu, 1987; Coe et al., 1987; and W. Lin et al., unpublished data, 1984]. These studies are derived from a wide area encompassing several geographic and tectonic provinces of the NCB: four of them from a redbed sequence of Late Permian age near Taiyuan (general location

TABLE 2. The Overprinted Component from Localities in Hebei and Shanxi Provinces

Locality	n/N	R/N	C	D	I	a95	k	Long	Lat	A95	K
Lincheng	7/13	3/4	G	2.2	57.0	11.6	28.2	180.8	88.0	15.9	15.4
			S	56.5	44.0	11.8	27.0	200.1	42.2	12.1	26.0
Kouquan	19/30	4/15	G	360.0	64.1	6.2	29.9	110.5	82.5	8.7	15.8
			S	354.7	36.7	6.5	28.0	308.7	70.5	6.0	32.6
Jingle	20/20	0/20	G	10.0	61.4	7.5	19.7	161.5	80.7	10.6	10.5
(West limb)			S	71.1	23.5	7.6	19.3	203.3	22.8	6.4	27.2
Jingle	36/36	0/36	G	0.0	51.4	4.3	31.6	295.2	84.4	5.3	21.2
(East limb)			S	333.6	43.2	4.3	31.6	359.7	64.5	4.9	25.1
Average for Jingle area											
	56/56	0/56	G	3.0	55.0	4.0	23.7	237.0	87.2	5.2	14.5
			S	10.4	47.5	10.7	4.1	258.5	79.4	12.5	3.3
Grand average N = 4			G	2.9	58.5	6.9	179.5	180.2	89.5	7.4	155.5
			S	23.5	45.9	42.0	5.8	227.5	69.5	51.8	4.0

n/N: number of samples exhibiting the overprint component/number of samples collected and measured;
R/N: number of reversed (R) to number of normal (N) samples;
C: coordinates, G = geographic, S = stratigraphic;
D: mean declination in degrees;
I: mean inclination in degrees;
a95: radius (degrees) of circle of 95% confidence about the mean direction;
k: Fisher precision parameter for mean direction;
Long: east longitude of Virtual Geomagnetic Pole (VGP);
Lat: north latitude of VGP;
A95: radius of circle of 95% confidence about the VGP;
K: Fisher precision parameter for the pole.

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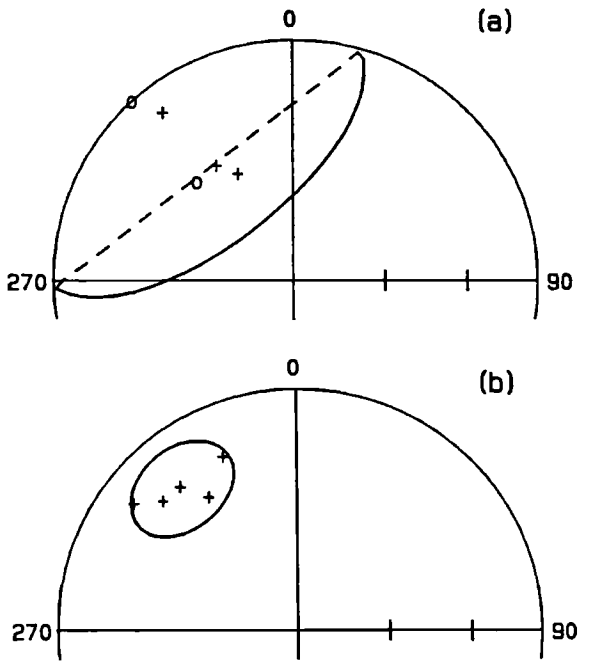


Fig. 7. Site mean directions of the characteristic component of the Late Permian rocks from the three regions on the NCB examined in this study; the data pass the regional fold test at more than 95% confidence, as indicated by a factor of 8.6 times increase in Fisher's precision parameter k (see Table 3); (a) directions plotted in geographic coordinates, and (b) stratigraphic coordinates. Open circles and dashed boundaries on confidence cones denote upper hemisphere projection, crosses and solid boundaries indicate lower hemisphere projection.

Lat = 37.8° N, Long = 112.4° E), Shanxi province, one from the western Beiqilianshan of Gansu province (Lat = 39.8° N, Long = 97.8° E), and one from Zheshiaobao (Lat = 42.3° N, Long = 110.2° E) of Inner Mongolia (Figure 1).

Among these Permian rock localities, the sections near Taiyuan in Shanxi are most extensively studied both geologically and paleomagnetically. McElhinny et al. [1981] first reported their results from the type section at Ximing (Lat = 37.8° N, Long = 112.3° E) where the Late Permian redbeds sequence is well developed. The beds are essentially flat-lying, so no fold test is applicable. Thermal demagnetization up to 670° C revealed the characteristic direction in about half of the samples, with both normal and reversed polarity. The polarity choice made first by McElhinny et al. [1981] has been confirmed by recent more detailed magnetostratigraphic studies [B. Embleton et al., unpublished data, 1987]. Subsequently, J. Lin et al. [1985], Zhang [1984], and W. Lin et al. [unpublished data, 1984] reported results from the same area. The results of the first two studies were derived from the same section and, not surprisingly, are almost identical to each other and virtually the same as McElhinny et al.'s [1981]. The results of the latter study are from two subsections (Qiligou section: Lat = 37.9° N, Long = 112.4° E; Duerping section: Lat = 37.9° N, Long = 112.3° E), southwest of Taiyuan [W. Lin et al., unpublished data, 1984]. The beds there are also flat-lying (with dips less than 10 degrees), but consist of both Late Permian redbeds and grey to yellow marl. The characteristic directions are reported to be carried by both magnetite and hematite, and agree well with others in this area. Application of the F-test of Watson [1965] indicates that these directions are indistinguishable at 95% confidence from each other and from our own.

In contrast to the studies in Taiyuan, the other two coeval results are derived from widely separated localities on the NCB. Preliminary results reported by Liu [1987] at a meeting in China are for Late Permian rocks from the Zheshiaobao area in Inner Mongolia (Lat = 42.3° N, Long = 110.2° E). Details on demagnetization work are not known, but the reported direction and corresponding pole position are in good agreement with those in Shanxi and Hebei provinces discussed above. The results reported by Coe et al. [1987] are very interesting. Their data are derived from coeval rocks some 1,300 km away to the west in the Dahuanggou section in Gansu province (Lat = 39.8 °N, Long = 97.8 ° E).

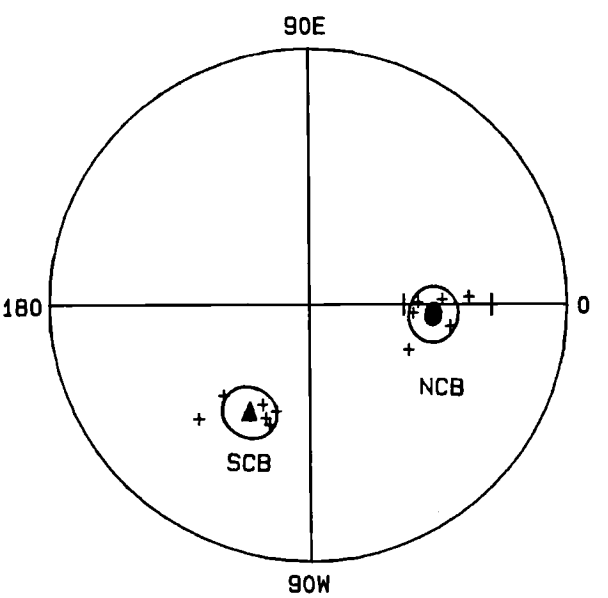


Fig. 8. Equal-area plot of Late Permian paleomagnetic poles with A95 confidence limits for the NCB and SCB; large triangle and solid circle represent means for the SCB and NCB, respectively.

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TABLE 3. The Characteristic Component from Four Sites in Hebei and Shanxi Provinces

Locality	n/N	R/N	DAM	C	D	I	a95	k	Long	Lat	A95	K
Lincheng	6/13	3/3	LSF	G	316.7	-0.9	12.4	30.0	351.1	35.0	9.6	49.8
				S	324.7	34.4	11.3	36.4	4.1	54.2	8.7	60.5
	12/13	7/5	CMB	G	317.9	-0.4	10.0	32.2	350.3	35.9	6.1	51.5
				S	326.3	34.3	6.9	42.0	1.4	55.2	6.3	48.2
Kouquan	7/29	4/3	LSF	G	330.6	39.9	8.9	47.2	358.1	59.9	8.9	47.4
				S	334.1	13.9	8.1	56.2	335.4	49.8	6.5	86.2
	19/29	11/8	CMB	G	332.5	48.8	9.1	14.8	7.0	65.3	10.2	11.9
				S	336.8	23.0	9.0	15.2	335.7	55.2	7.5	21.0
Jingle	7/20	4/3	LSF	G	318.3	-45.0	16.0	15.2	329.5	11.7	14.1	19.4
(West limb)				S	309.0	12.5	15.2	16.8	0.3	34.2	13.7	20.4
	10/20	6/4	CMB	G	315.4	-42.5	13.8	13.5	333.3	14.2	14.0	12.9
				S	307.7	15.7	13.3	14.5	3.2	34.1	10.6	21.7
Jingle	21/36	14/7	LSF	G	324.3	41.7	8.9	13.8	7.9	56.9	10.1	11.0
(East limb)				S	313.1	23.4	8.9	13.8	2.6	41.6	8.4	15.4
	23/36	16/7	CMB	G	326.1	42.2	8.4	13.9	5.8	58.0	8.6	13.3
				S	313.8	24.2	8.4	13.9	2.6	41.7	7.1	18.9
Average Jingle												
	33/56	22/11	CMB	G	322.8	23.5	16.1	3.8	353.3	47.2	11.8	6.3
				S	310.7	20.0	7.5	14.2	2.0	39.7	6.9	16.5
Grand average												
N = 4 sites			CMB	G	322.3	13.4	55.0	3.8	349.8	44.3	30.4	10.1
				S	320.9	24.7	16.3	32.8	356.9	47.1	15.5	35.9

DAM: data analyzing methods; LSF: least-squares-best-fit on line data [Kirschvink, 1980]; CMB: combined line and plane data analysis [McFadden and McElhinny, 1988];
Other abbreviations have the same meaning as Table 2.

The characteristic component is reversely magnetized (southeasterly and upward) and passes the fold test at greater than 95% confidence. The age of folding is uncertain but likely to be Early Triassic [Z. Meng, personal communication, 1987]. If this is true, the magnetization is almost certainly a record of the Late Permian geomagnetic field.

As shown in Table 3 and Figure 8, the virtual geomagnetic poles (VGPs) derived from directions of these areas are statistically indistinguishable from each other and agree well with our results. Thus, the data are reasonably representative of the whole North China Block.

Overall Discussion of Paleomagnetic Results from Both the NCB and SCB Blocks

The Permian paleomagnetic poles for the SCB and NCB computed from the paleomagnetic directions discussed above are distinct at greater than the 99% confidence level (Table 1 and Figure 8). Because the data for both the SCB and NCB pole positions pass fold and reversal tests and are mutually consistent, we believe they represent primary remanence with little or no contamination by unremoved secondary components. Thus, we must look for tectonic causes to explain the discrepancy. Moreover, these data are representative of large areas on the SCB and NCB (Figure 1), so that local rotation of small blocks is not a likely explanation.

The paleolatitudes implied by the data from the SCB and NCB are similar (near the equator), but the mean direction of paleo-north for the SCB is rotated clockwise more than 60 degrees with respect to that for the NCB. Thus, the NCB and SCB could not have been connected in their present configuration in Late Permian time. The class of single rotations capable of explaining the discrepancy in paleomagnetic results are about Euler poles lying on the great circle that is equidistant between the SCB and NCB paleomagnetic poles (Figure 9). There are, of course, any number of more complex solutions involving sequential rotations about more than one Euler pole. At this point, however, we choose to restrict our consideration to three of the simplest possible cases suggested by the geometrical relationships shown in Figure 9:

1. Rotation about an Euler pole at the north pole, about 60 degrees north of the NCB-SCB boundary;
2. Rotation about an Euler pole at the equator, about 30 degrees south of the NCB-SCB boundary;
3. Rotation about an Euler pole at the eastern end of the NCB-SCB boundary.

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Fig. 9. Map of mean paleomagnetic poles and 95 percent confidence circles for the NCB and SCB for Late Permian time. Heavy line near the 120 degree meridian is the bisecting great circle along which lie the Euler poles (numbered solid circles) corresponding to three cases considered in the text for the rotation of the SCB relative to the NCB. Dashed lines are hypothetical transform faults associated with Euler poles 1 and 2.

The first two cases represent alternative, simplified possibilities for ancient transform boundaries between the SCB and NCB: (1) a gigantic, east-west fault system (in today's geographic frame) and (2) a shorter fault system that curves around to the southwest, roughly parallel to the SCB boundary (Figure 9). Transform boundaries are preferred by those geologists who maintain that there has been no ocean-closing event between the SCB and NCB since mid-Paleozoic, and who cite structural and microstructural evidence for sinistral shear in the Qinling fold belt parallel to the NCB-SCB boundary (Zhang et al., 1984; Mattauer et al., 1985; Z. Q. Xu et al., 1986; Z. Q. Xu, 1987). Although the age of these shear zones was argued to be late Paleozoic (with Cenozoic reactivation), the possibility of early Mesozoic displacement apparently cannot be excluded (M. Mattauer, personal communication, 1988). In the first example, however, prohibitively large sinistral displacements are required to achieve the present separation of the SCB and NCB Late Permian paleomagnetic poles (70 percent of the circumference of the small circle that defines the transform, or 23,000 km!). Even if one takes liberty with the geological observations and assumes the opposite sense of shear, a dextral displacement of 11,000 km would be needed.

The second example is more plausible. About 4,300 km of sinistral displacement is required, which is only 15 percent of the circumference of the small circle corresponding to the transform fault. If one relaxes the paleomagnetic constraints slightly, allowing the Euler pole to lie a little west of the bisecting great circle at lat = 15 ^{o}N and long = 110 ^{o}E, the displacement around the smallest small circle that crudely fits the northern and western boundaries of the SCB can be cut to 2,500 km. It must be emphasized, however, that these models are completely untested geologically. At this moment we know of no evidence for such huge strike-slip displacements of Mesozoic age.

The third case entails subduction and/or compression along the NCB-SCB boundary, and is suggested by the fact that the bisecting great circle of Figure 9 intersects that boundary near its eastern end. A rotation of 67 degrees about an Euler pole at this point of intersection brings the SCB and NCB poles into coincidence. As shown in Figure 10, this is what might be expected if collision occurred initially at this point in easternmost China and progressed westward as the SCB rotated clockwise relative to the NCB. Of course, what counts is relative rotation, so one could say equally well that the NCB rotated counterclockwise relative to the SCB, or that they both rotated relative to an external reference such as the geographic poles. Although the paleomagnetic constraints can tell us nothing about the nature of the intervening crust and the mechanism by which it was consumed, this model is clearly more acceptable to those geologists who believe that a Permo-Triassic ocean existed between the SCB and NCB and that closed in Late Triassic or Jurassic time [Klimetz, 1983; Wang et al., 1983; Li et al., 1984; Ji and Coney, 1985; Sengor, 1985; Yang et al., 1986; J. W. Xu et al., 1987; Hsu et al., 1987].

The debates over whether this Paleotethys Ocean closed in the Triassic or the Devonian still continues, and stems in large part from a radical disagreement over the age of deposition of a flysch belt in the collision zone (see Z. Q. Xu, 1987 and Hsu et al., 1987 for advocacy of a Devonian or Triassic age assignment, respectively). Although we are not in a position to contribute directly to the solution of this disagreement, we can draw attention to reports on the distribution of marine sediments of indubitable Triassic age along the northern edge of the SCB that are remarkably consistent with the paleomagnetic model for rotation and collision given in case 3 above. These sediments are succeeded by Late Triassic and Jurassic terrestrial deposits, and appear to record the end of an extensive period of passive-margin sedimentation [Hsu et al., 1987]. They are well developed along the northwestern margin of the

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SCB, but thin eastward and are represented mainly by littoral and subaerial facies on the northeastern margin [Wang et al., 1982; Yang et al., 1982; Zhu et al., 1984]. According to the detailed field study of Zhu et al. [1984], the sea retreated from east to west starting in the Early Triassic. These geologic observations and inferences are readily explained if an ocean between the NCB and SCB had already closed in easternmost China due to collision, but remained open to the west until Late Triassic or Early Jurassic time, as envisioned in Figure 10.

Early Triassic data (Table 1) for the NCB and SCB are broadly similar to those for the Late Permian. Two studies [Lin, 1984 and Lin et al., 1985; Zhang, 1984] near Taiyuan in Shanxi province (Lat = 37.8 ^{o}N, Long = 112.4 ^{o} E) and one study [Tan, 1988] at two sections of the Ji and Jingle areas of Shanxi province (general location at Lat = 36.2 ^{o} N, Long = 110.8 ^{o} E, and Lat = 38.5 ^{o} N, Long = 112.1^{o} E, respectively) yield directions that agree well with those from the Late Permian rocks from the same general area. A fourth result [Lin, 1984 and Lin et al., 1985] from the Benxi area (Lat = 41.3 ^{o} N, Long = 123.7^{o} E), far to the east in Liaoning province, does not agree with the other two for the NCB, even after correction of a mistake in the calculation of the mean direction and pole (Table 1). The inclination is concordant with the other two, but the declination is rotated about 40 degrees counterclockwise. One possible interpretation is that the sampled area may have been rotated due to the nearby Tancheng-Lujiang Fault. The possibility of such local rotations near this major fault zone has been suggested by others [Courtillot and Besse, 1986; Kent et al., 1987]. For this reason, and because no evidence was presented demonstrating that the magnetization reported by Lin et al. [1985] is primary, we exclude this result from further consideration. On the SCB, seven Early and one Middle Triassic poles [Chan et al., 1984; Lin et al., 1985; Opdyke et al., 1986; Zhang et al., 1987; Heller et al., unpublished data, 1988] agree well with each other and are similar, although a little to the west, of the Late Permian poles. Because of their wide distribution (Figure 1), they further support the contention that the paleomagnetic results are representative of the whole SCB.

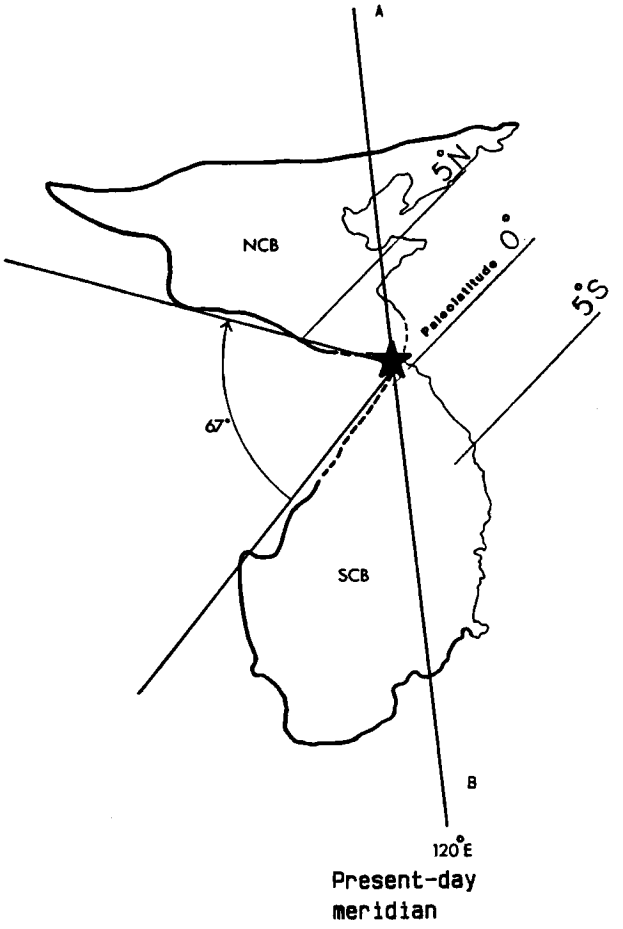


Fig. 10. Schematic reconstruction and paleolatitudes of the NCB and SCB in Permo-Triassic time. The NCB is arbitrarily kept in its present orientation relative to north. The boundaries are simplified in the collision zone, and the Tancheng-Lujiang fault is omitted. Line A-B is a portion of the bisecting great circle of Figure 9.

Thus, to a good approximation, the Early Triassic data are consistent with the same models discussed above in relation to the Late Permian data. If there were a representative number and distribution of Early Triassic poles for the NCB, it would have been preferable to base the models on the data for this time. Because no such poles exist, a better way to incorporate them is to average the two sets of data to obtain a Permo-Triassic mean pole position for each block. This has been done in Table 1. Changes to the earlier discussion caused by using these poles are minor, the main effect being a shift of the bisecting great circle of Figure 9 a few hundred kilometers to the west.

The question of when the SCB and NCB reached essentially their present relative positions still cannot be answered precisely. Well-documented studies show that the SCB and NCB paleomagnetic poles are statistically indistinguishable in Late Cretaceous time [Kent et al., 1986; Lee et al., 1987]. Therefore, large-scale (paleomagnetically detectable) relative motion between the two blocks was completed sometime between the Late Triassic and Late Cretaceous. Middle and Late Jurassic pole positions [Lin et al., 1985 and Lin, 1984] have been published that are similar for both blocks, but no documentation was offered to demonstrate that these rocks escaped the Late Cretaceous overprinting that is so pervasive in eastern China [Kent et al., 1987; Zhao, 1987]. We know

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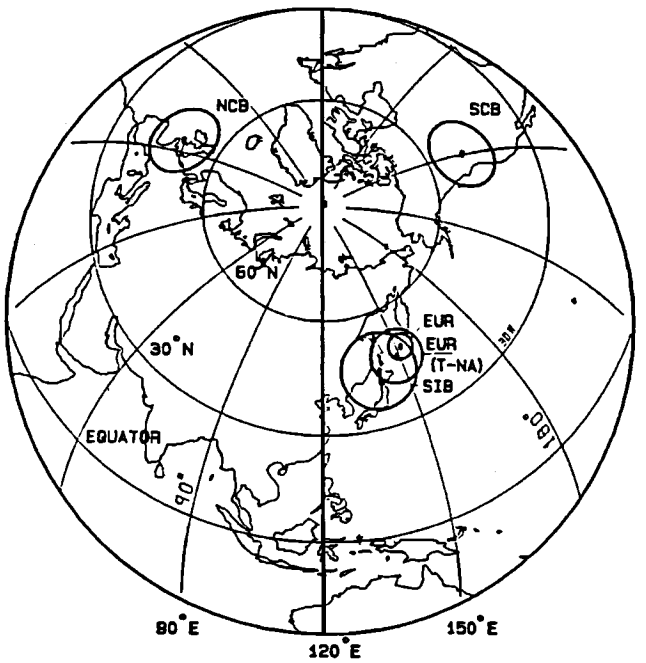


Fig. 11. Mean Late Permian paleomagnetic poles for NCB, SCB, Siberia, and Europe. NCB: Lat = 49.7 °N, Long = 355.9 °E, A95 = 7.2°, SCB: Lat = 50.0 °N, Long = 240.6 °E, A95 = 8.4°, Siberia (SIB): Lat = 45.0 °N, Long = 141.0 °E, A95 = 10.0° [Khramov, 1982], Europe (EUR): Lat = 48.9 ° N, Long = 151.9 °E, A95 = 3.3° [Westphal et al., 1986], and Europe transferred from North American APWP [EUR (T-NA)]: Lat = 47.0 ° N, Long = 149.0 °E, A95 = 7.0° [Westphal et al., 1986].

of no paleomagnetic data available for the Late Triassic and Early Jurassic. Because no Jurassic marine strata have been found in the collision zone [Yang et al., 1986], we currently favor the hypothesis that North and South China initially collided in the Early Triassic and had completed most of their relative rotation sometime in the Jurassic.

A final question concerns the time of amalgamation of China and Siberia. Almost all recent geological articles and books [Zhang et al., 1984; Li et al., 1984; Ji and Coney, 1986; Yang et al., 1986] and one paleomagnetic interpretation [Lin et al., 1985, p. 448] suggest that the NCB accreted to Siberia by Late Permian. The Late Permian and Early Triassic paleomagnetic results summarized in Table 1 and Figure 11 strongly contradict these interpretations. As pointed out first by McElhinny et al. [1981] and more recently and convincingly by Opdyke et al. [1986], the Permo-Triassic paleomagnetic data imply that both the SCB and the NCB lay several thousand kilometers farther south of Siberia compared to their present relative positions. This conclusion is also supported by recently published paleomagnetic results of Permian andesites and tuffites from three localities in northern and central Mongolia [Pruner, 1987], which imply that northern and central Mongolia (Lat = 48.1 - 49.5 °N, Long = 105 - 106 ° E) was probably a part of the NCB in the Permian, and was widely separated from the Siberian Block at that time. Moreover, one cannot easily escape this conclusion by claiming that the Siberian data may be in serious error. As shown in Figure 11, recently selected Late Permian poles for both Siberia and Europe agree reasonably well with each other and with the poles transferred from the North American APWP according to plate reconstruction models [Westphal et al., 1986].

How and when the relative displacement between China and Siberia was accommodated remains an unsolved problem of the first order in the tectonics of Asia. Late Cretaceous poles for the NCB, SCB, and the Siberian blocks overlap one another [Gurarii et al., 1966; Lee et al., 1987; Kent et al., 1986; Westphal et al., 1986; Khramov et al., 1985]. Thus, the major displacements between the three blocks were completed by the Late Cretaceous. Early Mesozoic paleomagnetic data from more northerly parts of the NCB are badly needed to help resolve the problem of the accretion of China to Siberia. Work on this problem on rocks from both Xinjiang autonomous region in the west and the Inner Mongolia autonomous region in the east is currently in progress at UCSC. Our own unpublished preliminary results together with other available data suggest the possibility that collision of the Chinese and Siberian blocks may have occurred initially in the late Paleozoic near the western end of an east-west boundary at about 49 ° N latitude and progressed eastward as the NCB rotated anticlockwise relative to Siberia during the Mesozoic.

Conclusion

Paleomagnetic evidence suggests that North and South China were not a stable unit in Permo-Triassic time. Collision and suturing probably occurred sometime within Triassic to Cretaceous time, and that this process was most likely completed during the Jurassic period. A simple model explaining the manner in which this may have taken place is suggested by the geometrical relationship of the NCB and SCB paleomagnetic poles. In this model, collision initially occurred in Early Triassic time at the eastern end of the SCB-NCB boundary and suturing progressed westward as the SCB rotated clockwise relative to the NCB. Eastward thinning and eventual disappearance of Triassic marine sediments on the northern margin of the SCB supports this hypothesis. An alternative simple model involves sinistral strike-slip movement on a fault system that trends westward within the Qinling fold belt and curves southwestward around the northwestern margin of the SCB. However,

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evidence for the large Mesozoic displacement (2,500 or more kilometers) required by this model is presently lacking. Accretion of the NCB to Siberia also took place sometime between Late Permian and Late Cretacous time, possibly by a model of collision and subsequent rotation similar to that suggested above for the SCB and NCB, but antithetic in sense of rotation. Mesozoic paleomagnetic data from all three blocks are needed to test and refine these ideas.

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